

# SPRINT 3 - Xavier Palet Bustillo

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***Revisión peer to peer: Natalia Cuenca***

**IT ACADEMY**

# Nivell 1

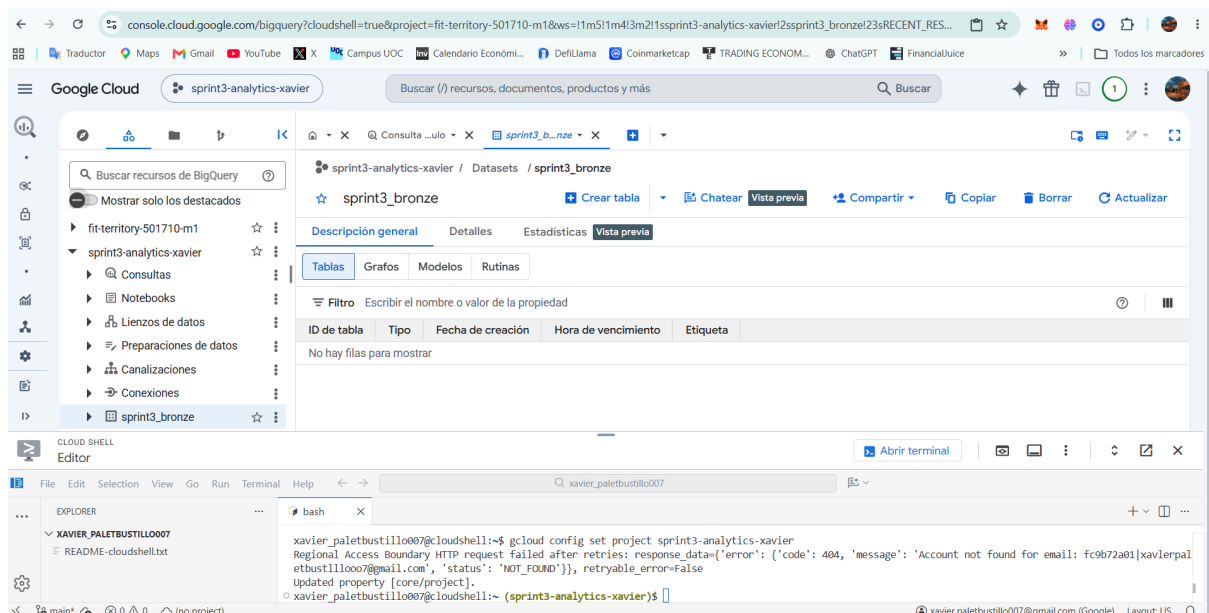
# Nivell 1

## Exercici 1 - Arquitectura de Dades (Lògica vs. Física)

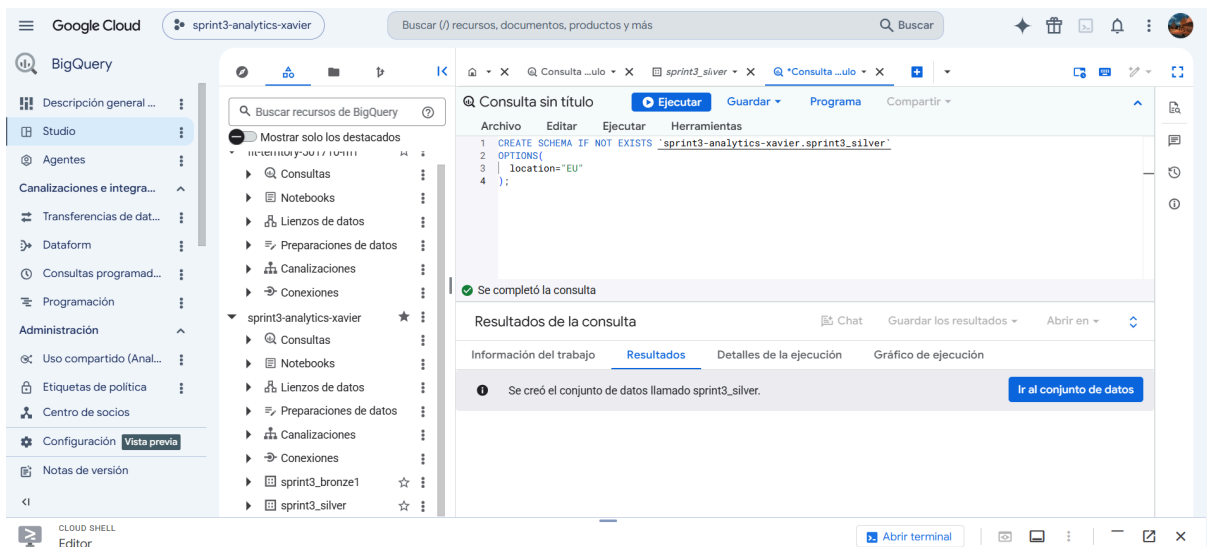
```
bash x

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to project-d5a5f7f6-fed0-4aa6-b83.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
xavier_paetbustillo007@cloudshell:~ (project-d5a5f7f6-fed0-4aa6-b83)$ gcloud projects create sprint3-analytics-xavier --name="Sprint 3 Analytics"
Regional Access Boundary HTTP request failed after retries: response_data={'error': {'code': 404, 'message': 'Account not found for email: fc9b72a01|xavlerpal
etbustill0007@gmail.com', 'status': 'NOT_FOUND'}}, retryable_error=False
Create in progress for [https://cloudresourcemanager.googleapis.com/v1/projects/sprint3-analytics-xavier].
Waiting for [operations/create_project.global.8731337004281970341] to finish...working.Regional Access Boundary HTTP request failed after retries: response_da
ta={'error': {'code': 404, 'message': 'Account not found for email: fc9b72a01|xavlerpaletbustill0007@gmail.com', 'status': 'NOT_FOUND'}}, retryable_error=Fals
e
Waiting for [operations/create_project.global.8731337004281970341] to finish...done.
Enabling service [cloudapis.googleapis.com] on project [sprint3-analytics-xavier]...
Regional Access Boundary HTTP request failed after retries: response_data={'error': {'code': 404, 'message': 'Account not found for email: fc9b72a01|xavlerpal
etbustill0007@gmail.com', 'status': 'NOT_FOUND'}}, retryable_error=False
Regional Access Boundary HTTP request failed after retries: response_data={'error': {'code': 404, 'message': 'Account not found for email: fc9b72a01|xavlerpal
etbustill0007@gmail.com', 'status': 'NOT_FOUND'}}, retryable_error=False
Regional Access Boundary HTTP request failed after retries: response_data={'error': {'code': 404, 'message': 'Account not found for email: fc9b72a01|xavlerpal
etbustill0007@gmail.com', 'status': 'NOT_FOUND'}}, retryable_error=False
Operation "operations/acet.p2-601860511332-3205b510-3ad9-49ca-9ee9-eb0956f0f98d" finished successfully.
xavier_paetbustillo007@cloudshell:~ (project-d5a5f7f6-fed0-4aa6-b83)$ gcloud config set project sprint3-analytics-xavier
Regional Access Boundary HTTP request failed after retries: response_data={'error': {'code': 404, 'message': 'Account not found for email: fc9b72a01|xavlerpal
etbustill0007@gmail.com', 'status': 'NOT_FOUND'}}, retryable_error=False
Updated property [core/project].
xavier_paetbustillo007@cloudshell:~ (sprint3-analytics-xavier)$
```

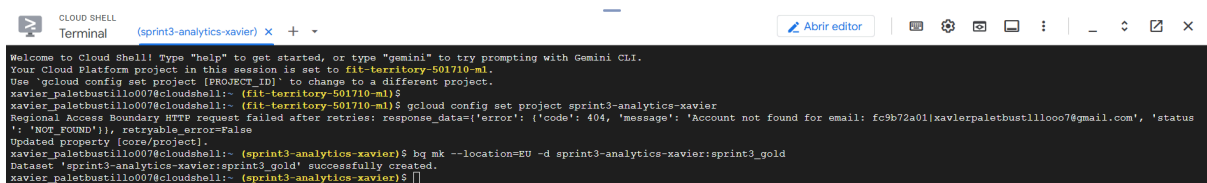
En la captura adjunta se demuestra la creación del proyecto *sprint3-analytics-xavier* mediante Google Cloud Shell (gcloud CLI).



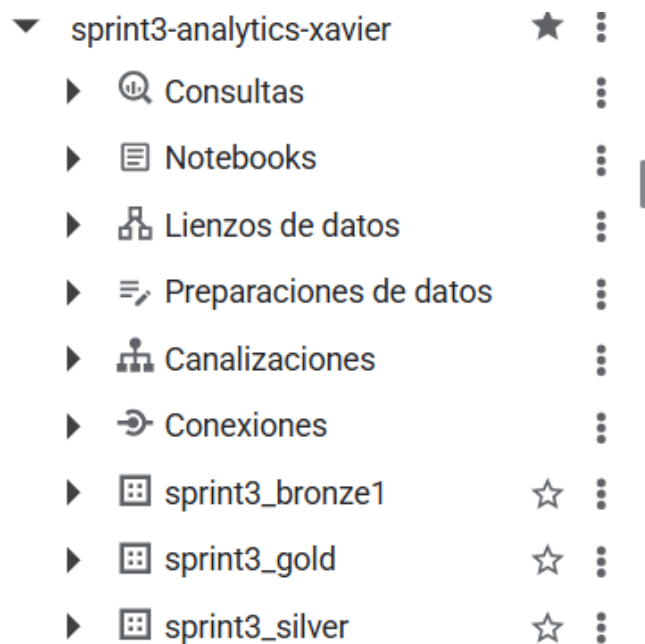
En este paso hemos creado el Dataset físico “bronce” en Google BigQuery utilizando la interfaz.



En este segundo paso hemos creado el Dataset físico “silver” utilizando código SQL “CREATE SCHEMA”.



Abrimos el terminal “Cloud SHELL” y escribimos el código “*bq mk --location=EU -d sprint3-analytics-xavier:sprint3\_gold*” para crear la última base de datos del ejercicio.



## Exercici 2: Ingesta en Capa Bronze (Connexió DDL)

```
1 CREATE OR REPLACE EXTERNAL TABLE `sprint3-analytics-xavi.sprint3_bronze.transactions_raw`  
2 OPTIONS (  
3   format = 'CSV',  
4   uris = ['gs://bootcamp-data-analytics-public/ERP/transactions.csv'],  
5   field_delimiter = ';',  
6   skip_leading_rows = 1  
7 );
```

The screenshot shows the BigQuery web interface. On the left, a sidebar lists various resources, with 'sprint3\_bronze' and its sub-table 'transactions\_raw' selected. The main panel displays the 'transactions\_raw' table schema. The schema table has columns: 'id' (STRING, NULLABLE), 'card\_id' (STRING, NULLABLE), 'business\_id' (STRING, NULLABLE), 'timestamp' (TIMESTAMP, NULLABLE), 'amount' (FLOAT, NULLABLE), 'declined' (INTEGER, NULLABLE), and 'product\_ids' (STRING, NULLABLE). Each column has a checkbox for selection and a description field. Below the schema table is a blue button labeled 'Editar esquema'.

Con el código “*CREATE TABLE...*” lo ejecutamos en una consulta en BigQuery para añadir la tabla “*transactions\_raw*” al dataset “*sprint3\_bronze*”.

- En Google SHELL buscamos los nombres de las columnas de la tabla “*companies\_raw*” con el siguiente comando para añadirlas al código para la creación de la tabla:

```
gsutil cat gs://bootcamp-data-analytics-public/ERP/companies.csv | head -n 5
```

The screenshot shows a Google Cloud Shell terminal window. The terminal displays the output of the command `gsutil cat gs://bootcamp-data-analytics-public/ERP/companies.csv | head -n 5`. The output shows the first five lines of a CSV file, including columns for company\_id, company name, phone, email, country, and website.

Q Consulta sin título Ejecutar Guardar Programa Compartir

Archivo Editar Ejecutar Herramientas

```
1 CREATE OR REPLACE EXTERNAL TABLE `sprint3-analytics-xavi.sprint3_bronze.companies_raw` (  
2   company_id STRING,  
3   company_name STRING,  
4   phone STRING,  
5   email STRING,  
6   country STRING,  
7   website STRING  
8 )  
9 OPTIONS (  
10   format = 'CSV',  
11   uris = ['gs://bootcamp-data-analytics-public/ERP/companies.csv'],  
12   skip_leading_rows = 1  
13 );
```

sprint3-analytics-xavi / Datasets / sprint3\_bronze / Tables / companies\_raw

☆ companies\_raw Consulta Chat Abrir en Compartir Borrar Actualizar

Esquema Detalles Estadísticas Linaje Perfil de datos Calidad de los datos

Filtro Escribir el nombre o valor de la propiedad ?

| <input type="checkbox"/> | Nombre del campo | Tipo   | Modo     | Descripción | Clave | Intercalación | Valor predeterminado | Etiquetas de políticas ? |
|--------------------------|------------------|--------|----------|-------------|-------|---------------|----------------------|--------------------------|
| <input type="checkbox"/> | company_id       | STRING | NULLABLE | -           | -     | -             | -                    | -                        |
| <input type="checkbox"/> | company_name     | STRING | NULLABLE | -           | -     | -             | -                    | -                        |
| <input type="checkbox"/> | phone            | STRING | NULLABLE | -           | -     | -             | -                    | -                        |
| <input type="checkbox"/> | email            | STRING | NULLABLE | -           | -     | -             | -                    | -                        |
| <input type="checkbox"/> | country          | STRING | NULLABLE | -           | -     | -             | -                    | -                        |
| <input type="checkbox"/> | website          | STRING | NULLABLE | -           | -     | -             | -                    | -                        |

- En Google SHELL buscamos los nombres de las columnas de la tabla “american\_users\_raw” con el siguiente comando para añadirlas al código para la creación de la tabla:

```
gsutil cat gs://bootcamp-data-analytics-public/CRM/american_users.csv | head -n 5
```

```
xavier_palaibustillo007@cloudshell:~ (fit-territory-501710-ml)$ gutil cat gs://bootcamp-data-analytics-public/CRM/american_users.csv | head -n 5
id,name,surname,phone,email,birth_date,country,city,postal_code,address
1,Zeus,Gamble,1-282-581-0551,interdum.enim@protonmail.edu,"Nov 17, 1985",United States,New York,10001,348-7818 Sagittis St.
2,Garrett,Mcconnell,(718) 257-2412,integer.vitae.nibh@protonmail.org,"Aug 23, 1992",United States,Philadelphia,19101,903 Sit Ave
3,Clara,Harrison,(522) 598-1365,interdum.feugiat@aol.org,"Apr 29, 1998",United States,Houston,77001,736-2863 Pellus St.
4,Howard,Stafford,1-411-740-3269,ornare.egestas@icloud.edu,"Feb 18, 1989",United States,Phoenix,85001,Ap #545-2244 Erat. Rd.
Exception ignored while flushing sys.stdout:
BrokenPipeError: [Errno 32] Broken pipe
```

Consulta sin título

Ejecutar

Guardar

Programa

Compartir

Archivo

Editar

Ejecutar

Herramientas

```

1 CREATE OR REPLACE EXTERNAL TABLE `sprint3-analytics-xavi.sprint3_bronze.american_users_raw` (
2   id STRING,
3   name STRING,
4   surname STRING,
5   phone STRING,
6   email STRING,
7   birth_date STRING,
8   country STRING,
9   city STRING,
10  postal_code STRING,
11  address STRING
12 )
13 OPTIONS (
14   format = 'CSV',
15   uris = ['gs://bootcamp-data-analytics-public/CRM/american_users.csv'],
16   skip_leading_rows = 1
17 );

```

Se completó la consulta

sprint3-analytics-xavi / Datasets / sprint3\_bronze / Tables / american\_users\_raw

american\_users\_raw

Consulta

Chat

Abrir en

Compartir

Borrar

Actualizar

Esquema

Detalles

Estadísticas

Linaje

Perfil de datos

Calidad de los datos

Filtro

Escribir el nombre o valor de la propiedad

| <input type="checkbox"/> | Nombre del campo | Tipo   | Modo     | Descripción | Clave | Intercalación | Valor predeterminado | Etiquetas de políticas |
|--------------------------|------------------|--------|----------|-------------|-------|---------------|----------------------|------------------------|
| <input type="checkbox"/> | id               | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | name             | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | surname          | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | phone            | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | email            | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | birth_date       | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | country          | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | city             | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | postal_code      | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | address          | STRING | NULLABLE | -           | -     | -             | -                    | -                      |

- Hacemos el mismo procedimiento para “*europaean\_users\_raw*”.



```
Consulta sin título Ejecutar Guardar Programa Compartir
Archivo Editar Ejecutar Herramientas
1 CREATE OR REPLACE EXTERNAL TABLE `sprint3-analytics-xavi.sprint3_bronze.european_users_raw` (
2   id STRING,
3   name STRING,
4   surname STRING,
5   phone STRING,
6   email STRING,
7   birth_date STRING,
8   country STRING,
9   city STRING,
10  postal_code STRING,
11  address STRING
12 )
13 OPTIONS (
14   format = 'CSV',
15   uris = ['gs://bootcamp-data-analytics-public/CRM/european_users.csv'],
16   skip_leading_rows = 1
17 );
```

Se completó la consulta

sprint3-analytics-xavi / Datasets / sprint3\_bronze / Tables / european\_users\_raw

☆ european\_users\_raw Consulta Chat Abrir en Compartir Borrar Actualizar

Esquema Detalles Estadísticas Linaje Perfil de datos Calidad de los datos

Filtro Escribir el nombre o valor de la propiedad

|                          | Nombre del campo | Tipo   | Modo     | Descripción | Clave | Intercalación | Valor predeterminado | Etiquetas de políticas |
|--------------------------|------------------|--------|----------|-------------|-------|---------------|----------------------|------------------------|
| <input type="checkbox"/> | id               | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | name             | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | surname          | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | phone            | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | email            | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | birth_date       | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | country          | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | city             | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | postal_code      | STRING | NULLABLE | -           | -     | -             | -                    | -                      |

- En Google SHELL buscamos los nombres de las columnas de la tabla “credit\_card\_raw” con el siguiente comando para añadirlas al código para la creación de la tabla:

`gsutil cat gs://bootcamp-data-analytics-public/CRM/credit_cards.csv | head -n 5`

```
xavier.paletbustillo007@cloudshell:~ (fit-territory-501710-ml)$ gsutil cat gs://bootcamp-data-analytics-public/CRM/credit_cards.csv | head -n 5
id,user_id,iban,pan,pin,cvv,track1,track2,expiring_date
ccU-2938,275,TR301950312213576817638661,5424465566813633,3257,984,1B8383712448554646*WovsxajDpwiev*8604114227,1B7653863056044187=800716333623,10/30/22
ccU-2945,274,DO26854763748537475216568689,5142423821948828,9080,887,1B4621311609958661*UftuyfsSeimxn*061062824127,1B4149568437843501=510714033021,08/24/23
ccU-2952,273,BG45IVQL52710525608255,4556 453 55 5287,4598,438,1B2183285104307501*CddyttcDxwfdg*590795543079,1B6778580257827162=6906859740027,06/29/21
ccU-2959,272,CR7242477244335841535,372461377349375,3583,667,1B7281111956795320*XocddijBckecd*0901625373,1B4246154489281853=280522391628,02/24/23
Exception ignored while flushing sys.stdout:
BrokenPipeError: [Errno 32] Broken pipe
```

Consulta sin título

Ejecutar Guardar Programa Compartir

Archivo Editar Ejecutar Herramientas

```
1 CREATE OR REPLACE EXTERNAL TABLE `sprint3-analytics-xavi.sprint3_bronze.credit_cards_raw` (  
2   id STRING,  
3   user_id STRING,  
4   iban STRING,  
5   pan STRING,  
6   pin STRING,  
7   cvv STRING,  
8   track1 STRING,  
9   track2 STRING,  
10  expiring_date STRING,  
11 )  
12 OPTIONS (  
13   format = 'CSV',  
14   uris = ['gs://bootcamp-data-analytics-public/CRM/credit_cards.csv'],  
15   skip_leading_rows = 1  
16 );
```

sprint3-analytics-xavi / Datasets / sprint3\_bronze / Tables / credit\_cards\_raw

credit\_cards\_raw Consulta Chat Abrir en Compartir Borrar Actualizar

Esquema Detalles Estadísticas Linaje Perfil de datos Calidad de los datos

Filtro Escribir el nombre o valor de la propiedad

| <input type="checkbox"/> | Nombre del campo | Tipo   | Modo     | Descripción | Clave | Intercalación | Valor predeterminado | Etiquetas de políticas |
|--------------------------|------------------|--------|----------|-------------|-------|---------------|----------------------|------------------------|
| <input type="checkbox"/> | id               | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | user_id          | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | iban             | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | pan              | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | pin              | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | cvv              | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | track1           | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | track2           | STRING | NULLABLE | -           | -     | -             | -                    | -                      |
| <input type="checkbox"/> | expiring_date    | STRING | NULLABLE | -           | -     | -             | -                    | -                      |

### Exercici 3: Càrrega de Dades Locals (Upload)

Buscar recursos de BigQuery

Mostrar solo los destacados

- Consultas
- Consultas compartidas
- Notebooks
- Lienzos de datos
- Preparaciones de datos
- Canalizaciones
- Conexiones
- sprint3\_bronze
  - american\_users\_raw
  - companies\_raw
  - credit\_cards\_raw
  - european\_users\_raw
  - products\_raw
  - transactions\_raw
- sprint3\_gold

sprint3-analytics-xavi / Datasets / sprint3\_bronze / Tables / products\_raw

products\_raw Consulta Abrir en Compartir Ver políticas de acceso de archivo

Esquema Detalles Vista previa Explorador de tablas Vista previa Estadísticas Linaje Perfil de datos

| <input type="checkbox"/> | Nombre del campo | Tipo    | Modo     | Descripción | Clave | Intercalación | Valor predeterminado |
|--------------------------|------------------|---------|----------|-------------|-------|---------------|----------------------|
| <input type="checkbox"/> | id               | INTEGER | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | product_name     | STRING  | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | price            | FLOAT   | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | colour           | STRING  | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | weight           | FLOAT   | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | warehouse_id     | STRING  | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | category         | STRING  | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | brand            | STRING  | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | cost             | FLOAT   | NULLABLE | -           | -     | -             | -                    |
| <input type="checkbox"/> | launch_date      | DATE    | NULLABLE | -           | -     | -             | -                    |

Editar esquema Ver políticas de acceso de archivo

## Exercici 4: Arquitectura i Rendiment. Materialització de Dades (Assistit per IA)

### a) Materialització de Dades (Assistit per IA):

- Prompt a Gemini y el código resultante:

The screenshot shows the Google Cloud Assist interface. At the top, there's a navigation bar with 'Cloud Assist' and a 'Vista previa' button. Below it, a warning message states: 'Gemini puede cometer errores. Verifica las respuestas y [usa el código con precaución](#).' The main content area contains a prompt in a light blue box: 'Write a SQL query to create a new table called transactions\_raw\_native in the sprint3\_bronze dataset. It should contain all data from the transactions\_raw table. Please use CREATE OR REPLACE TABLE so I don't get errors if I run it more than once.' Below the prompt, a status indicator says 'Finalizó el proceso de pensar'. The response is a SQL query displayed in a code editor:

```
CREATE OR REPLACE TABLE `sprint3-analytics-xavi.sprint3_bronze.transactions_raw_native` AS
SELECT
  *
FROM
  `sprint3-analytics-xavi.sprint3_bronze.transactions_raw`;
```

- Resultado de la tabla creada:

The screenshot shows the Google Cloud BigQuery interface. The left sidebar displays a project tree with 'sprint3\_bronze' expanded, showing various tables. The main panel shows the 'transactions\_raw\_native' table schema. The table has 10 columns, all with a 'NULLABLE' mode. The columns are: id (STRING), card\_id (STRING), business\_id (STRING), timestamp (TIMESTAMP), amount (FLOAT), declined (INTEGER), product\_ids (STRING), user\_id (INTEGER), lat (FLOAT), and longitude (FLOAT). The 'Descripción' column for all fields is empty.

| Nombre del campo | Tipo      | Modo     | Descripción | Clave | Intercalación | Valor predeterminado | Etiquetas de política |
|------------------|-----------|----------|-------------|-------|---------------|----------------------|-----------------------|
| id               | STRING    | NULLABLE | -           | -     | -             | -                    | -                     |
| card_id          | STRING    | NULLABLE | -           | -     | -             | -                    | -                     |
| business_id      | STRING    | NULLABLE | -           | -     | -             | -                    | -                     |
| timestamp        | TIMESTAMP | NULLABLE | -           | -     | -             | -                    | -                     |
| amount           | FLOAT     | NULLABLE | -           | -     | -             | -                    | -                     |
| declined         | INTEGER   | NULLABLE | -           | -     | -             | -                    | -                     |
| product_ids      | STRING    | NULLABLE | -           | -     | -             | -                    | -                     |
| user_id          | INTEGER   | NULLABLE | -           | -     | -             | -                    | -                     |
| lat              | FLOAT     | NULLABLE | -           | -     | -             | -                    | -                     |
| longitude        | FLOAT     | NULLABLE | -           | -     | -             | -                    | -                     |

### b) Auditoria de Costos:

- ¿Cuánto pesa la consulta a *transactions\_raw\_native*?

Consulta sin título [Ejecutar] [Guardar] Programa Compartir

Archivo Editar Ejecutar Herramientas

```
1 SELECT id FROM `sprint3-analytics-xavi.sprint3_bronze.transactions_raw_native`;
```

Se completó la consulta

Se está usando la cuota de procesamiento según demanda

Recientes < Resultados de la consulta [Chat] View all jobs Guardar los resultados ▾ Abrir en ▾

1:45 p.m. < Información del trabajo Resultados Visualización JSON Detalles de la ejecución Gráfico de e >

✓ SELECT id FROM `sprin...

|                        |                                |
|------------------------|--------------------------------|
| Hora de creación       | 9 jul 2026, 1:45:03 p.m. UTC+2 |
| Hora de inicio         | 9 jul 2026, 1:45:04 p.m. UTC+2 |
| Hora de finalización   | 9 jul 2026, 1:45:05 p.m. UTC+2 |
| Duración               | 1 s 45 ms                      |
| Bytes procesados       | 3.62 MB                        |
| Bytes facturados       | 10 MB                          |
| Milisegundos de ranura | 932                            |

Se puede ver que la consulta nativa procesa 3.62 MB y factura un mínimo 10 MB.

- ¿Cuánto pesa la consulta a *transactions\_raw*?

Consulta sin título [Ejecutar] [Guardar] Programa Compartir

Archivo Editar Ejecutar Herramientas

```
1 SELECT id FROM `sprint3-analytics-xavi.sprint3_bronze.transactions_raw`;
```

Se completó la consulta

Recientes < Resultados de la consulta [Chat] View all jobs Guardar los resultados ▾ Abrir en ▾

1:43 p.m. < Información del trabajo Resultados Visualización JSON Detalles de la ejecución Gráfico de e >

✓ SELECT id FROM `sprin...

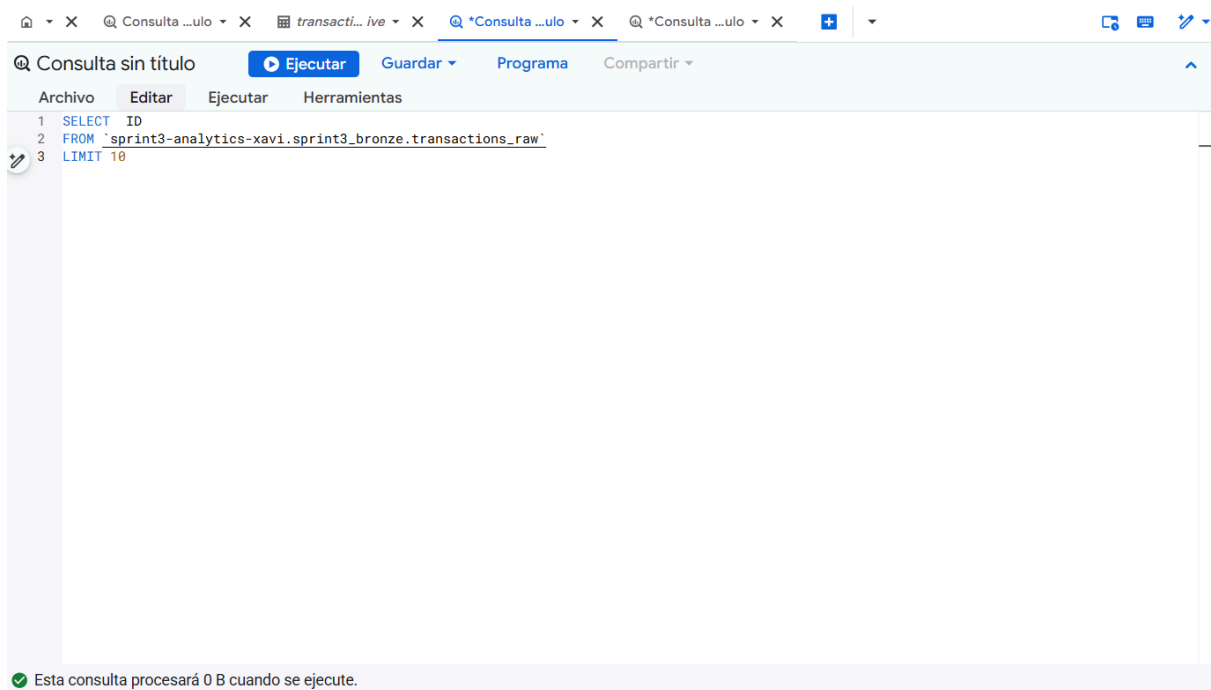
|                        |                                |
|------------------------|--------------------------------|
| Hora de creación       | 9 jul 2026, 1:43:25 p.m. UTC+2 |
| Hora de inicio         | 9 jul 2026, 1:43:25 p.m. UTC+2 |
| Hora de finalización   | 9 jul 2026, 1:43:27 p.m. UTC+2 |
| Duración               | 1 s 364 ms                     |
| Bytes procesados       | 12.61 MB                       |
| Bytes facturados       | 13 MB                          |
| Milisegundos de ranura | 1406                           |

Se puede ver que la consulta procesa 12.61 MB y factura un mínimo de 13 MB.

En resumen, se demuestra que el uso de tablas nativas reduce la cantidad de datos leídos, y por lo tanto es una forma más óptima y barata para las empresas con un gran volumen de datos.

c) El perill del LIMIT:

*Transactions\_raw* (consumirá 0 MB):

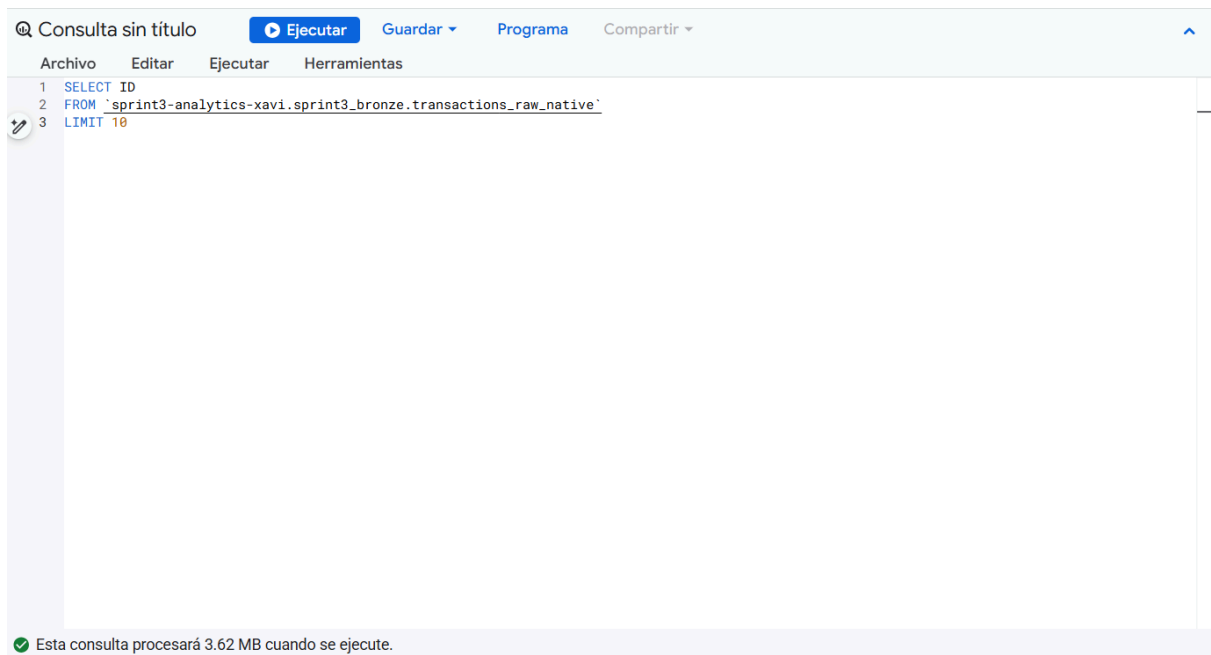


The screenshot shows a web-based SQL editor interface. At the top, there are several tabs, including one titled '\*Consulta ...ulo'. Below the tabs is a toolbar with buttons for 'Ejecutar' (Execute), 'Guardar' (Save), 'Programa', and 'Compartir' (Share). The main area contains a SQL query:

```
1 SELECT ID
2 FROM `sprint3-analytics-xavi.sprint3_bronze.transactions_raw`
3 LIMIT 10
```

At the bottom of the editor, a status bar displays a green checkmark and the text: 'Esta consulta procesará 0 B cuando se ejecute.'

*Transactions\_raw\_native* (consumirá 3.62 MB):



Cuando se ejecuta la consulta, el motor lee y escanea el 100% de los datos de la columna en toda la tabla, ya que la cláusula LIMIT 10 solo se aplica en la fase final de presentación para evitar saturar la pantalla del usuario con miles de filas.

### Exercici 5: Adaptació de Sintaxi (Reporting)

Consulta sin título Ejecutar Guardar Programa Compartir

Archivo Editar Ejecutar Herramientas

```
1 SELECT
2   CAST(timestamp AS DATE) AS fecha_limpia,
3   SUM(amount) AS ingresos_totales
4 FROM
5   `sprint3-analytics-xavi.sprint3-bronze.transactions_raw_native`
6 WHERE
7   EXTRACT(YEAR FROM CAST(timestamp AS DATE)) = 2021
8 GROUP BY
9   fecha_limpia
10 ORDER BY
11   ingresos_totales DESC
12 LIMIT 5;
```

Se completó la consulta

| Fila | fecha_limpia | ingresos_totales  |
|------|--------------|-------------------|
| 1    | 2021-10-27   | 11440.80999999... |
| 2    | 2021-01-22   | 11328.84000000... |
| 3    | 2021-02-11   | 11182.16          |
| 4    | 2021-04-29   | 11119.01          |
| 5    | 2021-03-27   | 10898.50999999... |

He utilizado la función “CAST(\_\_\_\_ AS DATE)” para transformar la columna *timestamp* en una fecha y así poder ejecutar las agrupaciones de fechas.

## Exercici 6: Consultes Complexes

```
1 -- Llista el nom, país i data de les transaccions realitzades per empreses que van fer operacions entre 100 i 200 euros en alguna
1 d'aquestes dates: 29-04-2015, 20-07-2018 o 13-03-2024 --
2
3 SELECT t.timestamp AS transaction_date, c.company_name, c.country
4 FROM `sprint3-analytics-xavi`.`sprint3_bronze`.`transactions_raw_native` AS t
5 JOIN `sprint3-analytics-xavi`.`sprint3_bronze`.`companies_raw` AS c
6 ON t.business_id = c.company_id
7 WHERE
8     t.amount BETWEEN 100 AND 200
9     AND DATE(t.timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
10 ORDER BY t.timestamp;
```

Resultados de la consulta

Chat

View all jobs

Guardar los resultados

Abrir en

| Información del trabajo |                         |                        |                | Resultados | Visualización | JSON | Detalles de la ejecución | Gráfico |
|-------------------------|-------------------------|------------------------|----------------|------------|---------------|------|--------------------------|---------|
| Fila                    | transaction_date        | company_name           | country        |            |               |      |                          |         |
| 1                       | 2015-04-29 06:37:02 UTC | Donec Fringilla PC     | France         |            |               |      |                          |         |
| 2                       | 2015-04-29 11:04:13 UTC | Auctor Mauris Vel LLP  | United States  |            |               |      |                          |         |
| 3                       | 2015-04-29 13:28:12 UTC | Non Magna LLC          | United Kingdom |            |               |      |                          |         |
| 4                       | 2015-04-29 13:46:12 UTC | Dictum Eu Corp.        | Canada         |            |               |      |                          |         |
| 5                       | 2015-04-29 13:53:21 UTC | Ac Libero Inc.         | United Kingdom |            |               |      |                          |         |
| 6                       | 2015-04-29 14:24:51 UTC | Amet Lorem LLP         | Spain          |            |               |      |                          |         |
| 7                       | 2018-07-20 00:32:35 UTC | Netus Et Malesuada Ltd | Netherlands    |            |               |      |                          |         |
| 8                       | 2018-07-20 02:40:13 UTC | Fusce Corp.            | United States  |            |               |      |                          |         |
| 9                       | 2018-07-20 05:47:38 UTC | Tempor Diam Institute  | Netherlands    |            |               |      |                          |         |

Resultados por página: 50 1 - 19 de 19



## Nivell 2

## Nivell 2

### Exercici 1 - Neteja de Productes (Data Quality)

Consulta sin título

EjecutarGuardarProgramaCompartir

ArchivoEditarEjecutarHerramientas

1

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16

CREATE OR REPLACE TABLE `sprint3-analytics-xavi.sprint3\_silver.products\_clean` AS

SELECT

id AS product\_id,

product\_name AS name,

price,

colour,

weight,

CAST(REPLACE(warehouse\_id, 'WH-', '') AS INT64) AS warehouse\_id,

category,

brand,

cost,

launch\_date

FROM

`sprint3-analytics-xavi.sprint3\_bronze.products\_raw`;

sprint3-analytics-xavi / Datasets / sprint3\_silver / Tables / products\_clean

products\_cleanConsultaChatAbrir enCompartirCopiarInstantánea

EsquemaDetallesVista previaExplorador de tablasVista previaEstadísticasLinajePerfil de datosCalidad de los datos

|                          | Nombre del campo | Tipo    | Modo     | Descripción | Clave | Intercalación | Valor predeterminado | Etiquetas de política |
|--------------------------|------------------|---------|----------|-------------|-------|---------------|----------------------|-----------------------|
| <input type="checkbox"/> | product_id       | INTEGER | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | name             | STRING  | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | price            | FLOAT   | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | colour           | STRING  | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | weight           | FLOAT   | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | warehouse_id     | INTEGER | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | category         | STRING  | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | brand            | STRING  | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | cost             | FLOAT   | NULLABLE | -           | -     | -             | -                    | -                     |
| <input type="checkbox"/> | launch_date      | DATE    | NULLABLE | -           | -     | -             | -                    | -                     |

### Exercici 2: Creació de Transaccions Netes (Capa Silver)

Consulta sin título

Ejecutar Guardar Programa Compartir

Archivo Editar Ejecutar Herramientas

```
1 CREATE OR REPLACE TABLE `sprint3-analytics-xavi.sprint3_silver.transactions_clean` AS
2 SELECT
3     id AS transaction_id,
4     card_id,
5     business_id,
6     timestamp,
7     IFNULL(SAFE_CAST(amount AS FLOAT64), 0) AS amount,
8     declined,
9     ARRAY(SELECT CAST(id AS INT64) FROM UNNEST(SPLIT(product_ids, ',')) AS id) AS product_ids,
10    user_id,
11    SAFE_CAST(lat AS FLOAT64) AS lat,
12    SAFE_CAST(longitude AS FLOAT64) AS longitude,
13
14 FROM `sprint3-analytics-xavi.sprint3_bronze.transactions_raw`;
```

### Exercici 3: Unificació d'Usuaris (UNION)

```

1 CREATE OR REPLACE TABLE `sprint3-analytics-xavi.sprint3_silver.users_combined` AS
2
3 SELECT
4     SAFE_CAST(id AS INT64) AS user_id,
5     name,
6     surname,
7     phone,
8     email,
9     birth_date,
10    country,
11    city,
12    postal_code,
13    address,
14    'EUA' AS origin
15 FROM `sprint3-analytics-xavi.sprint3_bronze.american_users_raw`
16
17 UNION ALL
18
19 SELECT
20     SAFE_CAST(id AS INT64) AS user_id,
21     name,
22     surname,
23     phone,
24     email,
25     birth_date,
26     country,
27     city,
28     postal_code,
29     address,
30     'Europa' AS origin
31 FROM `sprint3-analytics-xavi.sprint3_bronze.european_users_raw`;

```

sprint3-analytics-xavi / Datasets / sprint3\_silver / Tables / users\_combined

☆ users\_combined 🔍 Consulta 💬 Chat Abrir en ▾ 👤 Compartir ▾

Esquema Detalles Vista previa Explorador de tablas **Vista previa** Estadísticas

☰ Filtro Escribir el nombre o valor de la propiedad

| <input type="checkbox"/> | Nombre del campo | Tipo    | Modo     | Descripción | Clave | Int |
|--------------------------|------------------|---------|----------|-------------|-------|-----|
| <input type="checkbox"/> | user_id          | INTEGER | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | name             | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | surname          | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | phone            | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | email            | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | birth_date       | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | country          | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | city             | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | postal_code      | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | address          | STRING  | NULLABLE | -           | -     | -   |
| <input type="checkbox"/> | origin           | STRING  | NULLABLE | -           | -     | -   |

#### Exercici 4: Materialització de Companyies i Targetes de Crèdit

Creación de “*sprint3\_silver.companies\_clean*”:

```
1 CREATE OR REPLACE TABLE `sprint3-analytics-xavi.sprint3_silver.companies_clean` AS
2 SELECT
3     company_id,
4     company_name,
5     phone,
6     email,
7     country,
8     website,
9 FROM
10 `sprint3-analytics-xavi.sprint3_bronze.companies_raw`;
```

sprint3-analytics-xavi / Datasets / sprint3\_silver / Tables / companies\_clean

☆ companies\_cl... [Consulta](#) [Chat](#) [Abrir en ▾](#) [Compa](#)

Esquema Detalles Vista previa Explorador de tablas **Vista previa** Est

Filtro Escribir el nombre o valor de la propiedad

| <input type="checkbox"/> | Nombre del campo | Tipo   | Modo     | Descripción | Clave |
|--------------------------|------------------|--------|----------|-------------|-------|
| <input type="checkbox"/> | company_id       | STRING | NULLABLE | -           | -     |
| <input type="checkbox"/> | company_name     | STRING | NULLABLE | -           | -     |
| <input type="checkbox"/> | phone            | STRING | NULLABLE | -           | -     |
| <input type="checkbox"/> | email            | STRING | NULLABLE | -           | -     |
| <input type="checkbox"/> | country          | STRING | NULLABLE | -           | -     |
| <input type="checkbox"/> | website          | STRING | NULLABLE | -           | -     |

Creación de “*sprint3\_silver.credit\_cards\_clean*”:

```
1 CREATE OR REPLACE TABLE `sprint3-analytics-xavi.sprint3_silver.credit_cards_clean` AS
2 SELECT
3   id AS card_id,
4   user_id,
5   REPLACE(iban, ' ', '') AS iban,
6   REPLACE(pan, ' ', '') AS pan,
7   pin,
8   cvv,
9   track1,
10  track2,
11  expiring_date,
12 FROM
13  `sprint3-analytics-xavi.sprint3_bronze.credit_cards_raw`;
```

| Esquema                  | Detalles         | Vista previa | Explorador de tablas | Vista previa | Est   |
|--------------------------|------------------|--------------|----------------------|--------------|-------|
| <input type="checkbox"/> | Nombre del campo | Tipo         | Modo                 | Descripción  | Clave |
| <input type="checkbox"/> | card_id          | STRING       | NULLABLE             | -            | -     |
| <input type="checkbox"/> | user_id          | STRING       | NULLABLE             | -            | -     |
| <input type="checkbox"/> | iban             | STRING       | NULLABLE             | -            | -     |
| <input type="checkbox"/> | pan              | STRING       | NULLABLE             | -            | -     |
| <input type="checkbox"/> | pin              | STRING       | NULLABLE             | -            | -     |
| <input type="checkbox"/> | cvv              | STRING       | NULLABLE             | -            | -     |
| <input type="checkbox"/> | track1           | STRING       | NULLABLE             | -            | -     |
| <input type="checkbox"/> | track2           | STRING       | NULLABLE             | -            | -     |
| <input type="checkbox"/> | expiring_date    | STRING       | NULLABLE             | -            | -     |

## Nivell 3

## Nivell 3

### Exercici 1: La Vista de Màrqueting (Lògica de Negoci)

- Creamos la vista:



The screenshot shows a SQL query editor with a toolbar at the top containing buttons for 'Ejecutar' (Execute), 'Guardar' (Save), 'Programa', and 'Compartir' (Share). Below the toolbar is a menu bar with 'Archivo', 'Editar', 'Ejecutar', and 'Herramientas'. The main area displays a SQL query with line numbers 1 through 27. The query is a CREATE OR REPLACE VIEW statement for a view named 'v\_marketing\_kpis' in the schema 'sprint3-analytics-xavi.sprint3\_gold'. The view selects company information and calculates an average purchase amount, categorizing clients into 'Premium' or 'Standard' tiers based on their average purchase amount. The view is created by joining the 'companies\_clean' table with the 'transactions\_clean' table on the 'business\_id' field. The final SELECT statement orders the results by 'client\_tier' in ascending order and 'avg\_purchase\_amount' in descending order.

```
1 CREATE OR REPLACE VIEW `sprint3-analytics-xavi.sprint3_gold.v_marketing_kpis` AS
2 SELECT
3     c.company_name,
4     c.phone,
5     c.country,
6     ROUND(AVG(t.amount),2) AS avg_purchase_amount,
7     CASE
8         WHEN AVG(t.amount) > 260 THEN 'Premium'
9         ELSE 'Standard'
10    END AS Client_tier
11 FROM
12     `sprint3-analytics-xavi.sprint3_silver.companies_clean` c
13 JOIN
14     `sprint3-analytics-xavi.sprint3_silver.transactions_clean` t
15 ON
16     c.company_id = t.business_id
17 GROUP BY
18     c.company_name,
19     c.phone,
20     c.country;
21
22 SELECT *
23 FROM
24     `sprint3-analytics-xavi.sprint3_gold.v_marketing_kpis`
25 ORDER BY
26     client_tier ASC,
27     avg_purchase_amount DESC;
```



| <div>  <a href="#">sprint3-analytics-xavi</a> / <a href="#">Datasets</a> / <a href="#">sprint3_gold</a> / <a href="#">Tables</a> / <a href="#">v_marketing_kpis</a> </div>                                                                                                                                                                                                                                                                                               |                     |        |          |             |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|----------|-------------|-------|
| <div> <div>  <a href="#">v_marketing_kpis</a> </div> <div>  <a href="#">Consulta</a> </div> <div>  <a href="#">Chat</a> </div> <div> <a href="#">Abrir en</a> </div> <div>  <a href="#">Compartir</a> </div> </div> |                     |        |          |             |       |
| <div> <a href="#">Esquema</a> <a href="#">Detalles</a> <a href="#">Explorador de tablas</a> <a href="#">Vista previa</a> <a href="#">Estadísticas</a> <a href="#">Linaje</a> </div>                                                                                                                                                                                                                                                                                                                                                                       |                     |        |          |             |       |
| <div>  <b>Filtro</b> Escribir el nombre o valor de la propiedad </div>                                                                                                                                                                                                                                                                                                                                                                                                   |                     |        |          |             |       |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Nombre del campo    | Tipo   | Modo     | Descripción | Clave |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | company_name        | STRING | NULLABLE | -           | -     |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | phone               | STRING | NULLABLE | -           | -     |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | country             | STRING | NULLABLE | -           | -     |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | avg_purchase_amount | FLOAT  | NULLABLE | -           | -     |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Client_tier         | STRING | NULLABLE | -           | -     |

- Creamos la consulta personalizada para el departamento de marketing:

Consulta sin título

Ejecutar

Guardar

Programa

Archivo

Editar

Ejecutar

Herramientas

1

2

3

4

5

6

SELECT \*

FROM

sprint3-analytics-xavi.sprint3\_gold.v\_marketing\_kpis

ORDER BY

client\_tier ASC,

avg\_purchase\_amount DESC;

Resultados de la consulta

ChatView all jobsGuardar

| Información del trabajo | Resultados                  | Visualización  | JSON          | Detalles de la ejecución | Gráfico de ejecución |
|-------------------------|-----------------------------|----------------|---------------|--------------------------|----------------------|
| Fila                    | company_name                | phone          | country       | avg_purchase_a...        | Client_tier          |
| 1                       | Ac Fermentum Incorporated   | 06 85 56 52 33 | Germany       | 284.87                   | Premium              |
| 2                       | Pretium Neque Corp.         | 07 77 48 55 28 | Australia     | 276.16                   | Premium              |
| 3                       | Urna Convallis Associates   | 06 01 24 77 04 | United States | 274.24                   | Premium              |
| 4                       | At Associates               | 09 56 61 10 65 | New Zealand   | 272.21                   | Premium              |
| 5                       | Metus Vitae Associates      | 08 25 44 40 66 | Australia     | 270.08                   | Premium              |
| 6                       | Aliquet Diam Limited        | 02 76 61 47 46 | United States | 269.6                    | Premium              |
| 7                       | Nec Luctus LLC              | 02 14 71 75 73 | Norway        | 268.6                    | Premium              |
| 8                       | Neque Tellus Incorporated   | 04 43 18 34 19 | Ireland       | 267.85                   | Premium              |
| 9                       | Tortor Nunc Commodo Company | 05 35 92 77 16 | United States | 267.84                   | Premium              |
| 10                      | Cras Consulting             | 07 50 10 85 63 | Belgium       | 267.44                   | Premium              |
| 11                      | Eget Ipsum Ltd              | 03 67 44 56 72 | United States | 267.06                   | Premium              |
| 12                      | Sed LLC                     | 01 63 16 26 52 | Belgium       | 266.07                   | Premium              |
| 13                      | Amet Institute              | 06 33 40 21 33 | Australia     | 265.54                   | Premium              |
| 14                      | Pede Ultrices Ltd           | 04 52 40 26 17 | Belgium       | 265.36                   | Premium              |
| 15                      | Fusce Corp.                 | 08 14 97 58 85 | United States | 265.23                   | Premium              |
| 16                      | Tincidunt Quis Limited      | 01 78 18 81 44 | Norway        | 265.1                    | Premium              |

Exercici 2: Rànquing de Productes (La Potència dels Arrays)

Consulta sin título

Ejecutar Guardar Programa Compartir

Archivo Editar Ejecutar Herramientas

```
1 CREATE OR REPLACE TABLE `sprint3-analytics-xavi.sprint3_gold.product_sales_ranking` AS
2 WITH flattened_transactions AS (
3     SELECT
4         transaction_id,
5         id_producte_venut
6     FROM
7         `sprint3-analytics-xavi.sprint3_silver.transactions_clean`,
8         UNNEST(product_ids) AS id_producte_venut
9     SELECT
10        p.product_id,
11        p.name,
12        p.price,
13        p.colour,
14        COUNT(ft.id_producte_venut) AS total_sold
15    FROM
16        `sprint3-analytics-xavi.sprint3_silver.products_clean` p
17    LEFT JOIN
18        flattened_transactions ft
19    ON
20        p.product_id = ft.id_producte_venut
21    GROUP BY
22        p.product_id,
23        p.name,
24        p.price,
25        p.colour
26    ORDER BY
27        total_sold DESC;
```

| Fila | product_id | name                 | price  | colour  | total_sold |
|------|------------|----------------------|--------|---------|------------|
| 1    | 52         | riverlands the duel  | 31.52  | #727272 | 2654       |
| 2    | 29         | Tully maester Tarly  | 167.2  | #111111 | 2635       |
| 3    | 21         | duel Direwolf        | 96.9   | #e2e2e2 | 2609       |
| 4    | 16         | the duel warden      | 180.91 | #666666 | 2608       |
| 5    | 66         | mustafar jinn        | 2.12   | #545454 | 2601       |
| 6    | 87         | sith Jade            | 96.26  | #e2e2e2 | 2598       |
| 7    | 48         | rock Renly in        | 21.53  | #9e9e9e | 2597       |
| 8    | 33         | duel warden          | 127.09 | #191919 | 2597       |
| 9    | 23         | riverlands north     | 169.96 | #545454 | 2593       |
| 10   | 68         | Stark Karstark       | 167.37 | #333333 | 2589       |
| 11   | 88         | Stannis warden south | 167.15 | #515151 | 2587       |
| 12   | 28         | chewbacca mustafar   | 127.44 | #efefef | 2584       |
| 13   | 4          | warden south duel    | 71.89  | #111111 | 2584       |
| 14   | 73         | Dorne bastard        | 114.09 | #848484 | 2584       |
| 15   | 31         | Lannister            | 85.02  | #3f3f3f | 2580       |
| 16   | 34         | of north             | 128.48 | #919191 | 2580       |

### Exercici 3: Exportació de Resultats

En el resultado de la consulta anterior le damos a “*Guardar los resultados*” y cuando se despliega la pestaña le damos a “*Descarga local - CSV*”:

Resultados de la consulta

Chat

View all jobs

Guardar los resultados

Abrir en

Información del trabajo

Resultados

Visualización

JSON

Detalles de la ejecución

Gráfico de ejecución

| Fila | product_id | name                         | price  | colour  | total_sold |
|------|------------|------------------------------|--------|---------|------------|
| 85   | 94         | Direwolf stannis             | 6.93   | #c9c9c9 | 2480       |
| 86   | 94         | in south Winterfell          | 51.94  | #3a3a3a | 2479       |
| 87   | 45         | south                        | 92.65  | #bfbfbf | 2478       |
| 88   | 81         | the duel                     | 15.87  | #7f7f7f | 2469       |
| 89   | 30         | Karstark warden              | 79.53  | #606060 | 2468       |
| 90   | 1          | Direwolf Stannis             | 161.11 | #7c7c7c | 2468       |
| 91   | 51         | north Tully                  | 30.14  | #262626 | 2466       |
| 92   | 77         | Tarly Littlefinger           | 150.5  | #f4f4f4 | 2462       |
| 93   | 71         | Tully Dorne                  | 103.73 | #424242 | 2462       |
| 94   | 37         | Direwolf Littlefinger        | 26.66  | #aaaaaa | 2461       |
| 95   | 75         | giantsblood maester Karstark | 9.46   | #f9f9f9 | 2461       |
| 96   | 69         | Littlefinger the giantsblood | 57.25  | #b7b7b7 | 2457       |
| 97   | 19         | dooku solo                   | 60.33  | #3f3f3f | 2449       |
| 98   | 22         | chewbacca mustafar           | 150.02 | #cfcfcf | 2448       |
| 99   | 42         | sith                         | 83.51  | #494949 | 2425       |
| 100  | 43         | duel                         | 59.8   | #5b5b5b | 2414       |

Resultados por página:

Copiar en el portapapeles hasta 1 MB

Google Drive

CSV hasta 1 GB

JSONL (delimitado por líneas nuevas) hasta 1 GB

Hojas de cálculo de Google hasta 10 MB

Descarga local

CSV hasta 10 MB

JSON hasta 10 MB

Google Cloud

Cloud Storage

Tabla de BigQuery

Al abrir el documento CSV. a Google Sheets hay 4 celdas en la columna de price que me salen en formato de fecha. Parece que, como el separador de decimales es un punto (.) y mi Google Sheets tiene la configuración de España/UE donde se separa por coma (,) lo confunde y lo automatiza poniendo fechas.

Para solucionar esto con estas 4 celdas he tenido que cambiar la configuración del Google Sheets a Reino Unido y he cambiado las 4 celdas de formato a Números.

Product\_sales\_ranking

Archivo Editar Ver Insertar Formato Datos Herramientas Gemini Extensiones Ayuda

100% 123 Predet... 10 B I A

O23 fx

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Resume estos datos

|    | A          | B                 | C      | D       | E          | F | G | H | I | J | K | L | M | N | O |
|----|------------|-------------------|--------|---------|------------|---|---|---|---|---|---|---|---|---|---|
| 1  | product_id | name              | price  | colour  | total_sold |   |   |   |   |   |   |   |   |   |   |
| 2  | 52         | riverlands the du | 31.52  | #727272 | 2654       |   |   |   |   |   |   |   |   |   |   |
| 3  | 29         | Tully maester Ta  | 167.2  | #111111 | 2635       |   |   |   |   |   |   |   |   |   |   |
| 4  | 21         | duel Direwolf     | 96.9   | #e2e2e2 | 2609       |   |   |   |   |   |   |   |   |   |   |
| 5  | 16         | the duel warden   | 180.91 | #666666 | 2608       |   |   |   |   |   |   |   |   |   |   |
| 6  | 66         | mustafar jinn     | 2.12   | #545454 | 2601       |   |   |   |   |   |   |   |   |   |   |
| 7  | 87         | sith Jade         | 96.26  | #e2e2e2 | 2598       |   |   |   |   |   |   |   |   |   |   |
| 8  | 48         | rock Renly in     | 21.53  | #9e9e9e | 2597       |   |   |   |   |   |   |   |   |   |   |
| 9  | 33         | duel warden       | 127.09 | #191919 | 2597       |   |   |   |   |   |   |   |   |   |   |
| 10 | 23         | riverlands north  | 169.96 | #545454 | 2593       |   |   |   |   |   |   |   |   |   |   |
| 11 | 68         | Stark Karstark    | 167.37 | #333333 | 2589       |   |   |   |   |   |   |   |   |   |   |
| 12 | 88         | Stannis warden    | 167.15 | #515151 | 2587       |   |   |   |   |   |   |   |   |   |   |
| 13 | 28         | chewbacca musi    | 127.44 | #efefef | 2584       |   |   |   |   |   |   |   |   |   |   |
| 14 | 4          | warden south du   | 71.89  | #111111 | 2584       |   |   |   |   |   |   |   |   |   |   |
| 15 | 73         | Dorne bastard     | 114.09 | #848484 | 2584       |   |   |   |   |   |   |   |   |   |   |
| 16 | 31         | Lannister         | 85.02  | #3f3f3f | 2580       |   |   |   |   |   |   |   |   |   |   |
| 17 | 34         | of north          | 128.48 | #919191 | 2580       |   |   |   |   |   |   |   |   |   |   |
| 18 | 38         | ewok sith         | 55.43  | #666666 | 2578       |   |   |   |   |   |   |   |   |   |   |
| 19 | 36         | Lannister         | 96.22  | #a5a5a5 | 2575       |   |   |   |   |   |   |   |   |   |   |
| 20 | 61         | Winterfell Lannis | 28.01  | #848484 | 2575       |   |   |   |   |   |   |   |   |   |   |
| 21 | 54         | dooku solo        | 13.14  | #e5e5e5 | 2575       |   |   |   |   |   |   |   |   |   |   |
| 22 | 24         | south duel tourm  | 48.99  | #aaaaaa | 2574       |   |   |   |   |   |   |   |   |   |   |
| 23 | 11         | Karstark Dorne    | 49.7   | #141414 | 2573       |   |   |   |   |   |   |   |   |   |   |

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